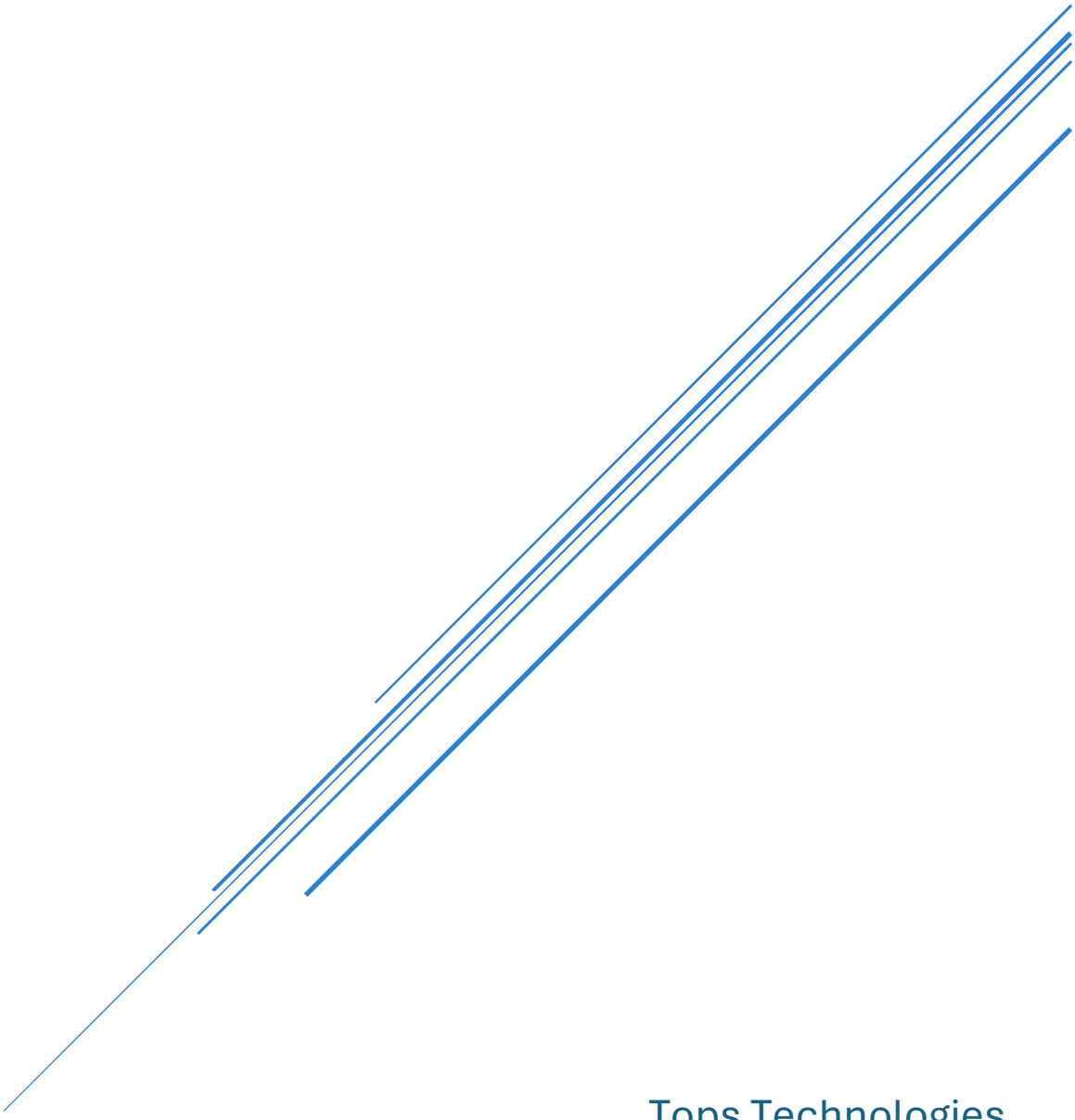
NETWORK CABLE AND CONNECTION

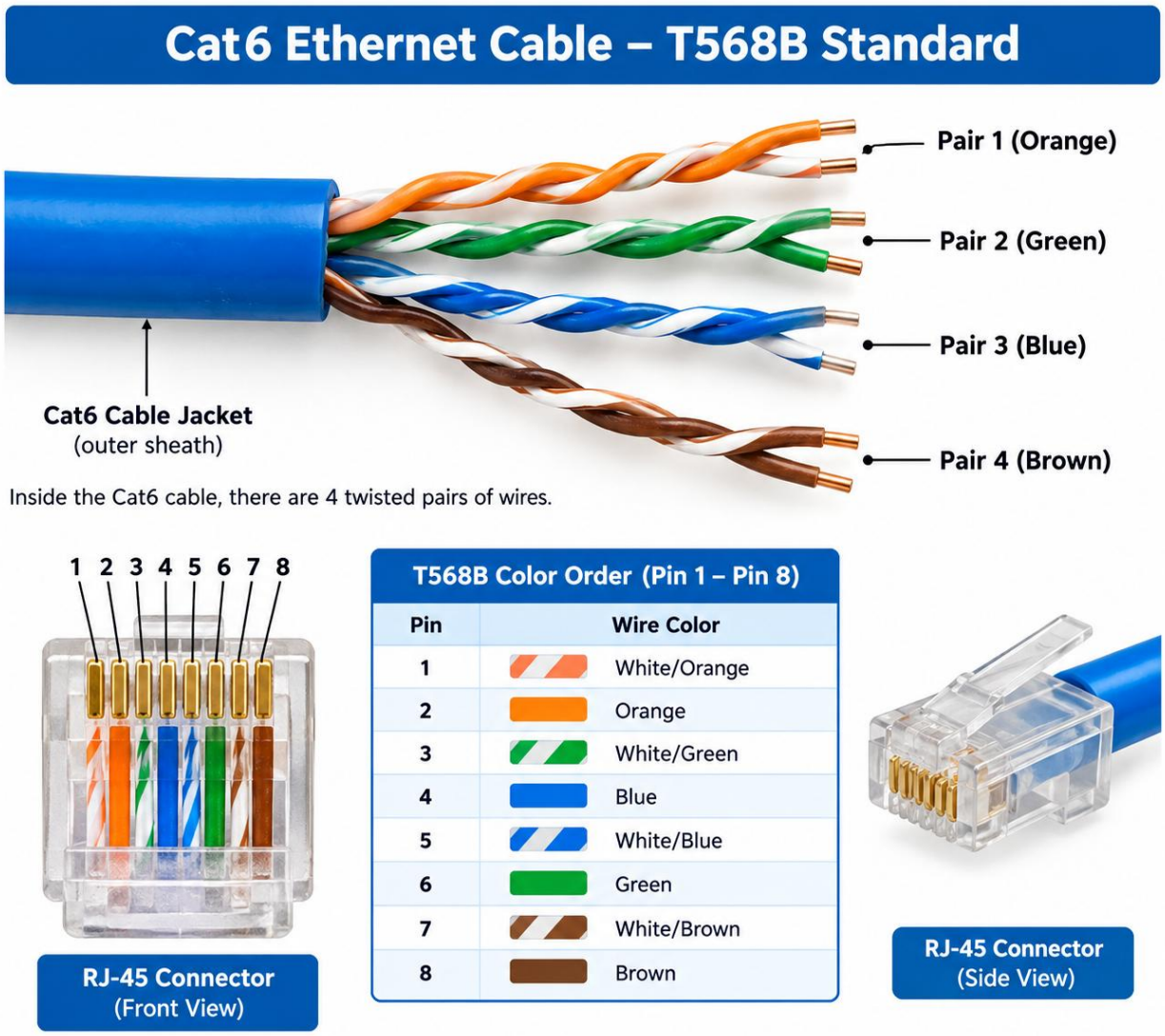
Assignment - 4



Tops Technologies
Cyber Security

1. **Draw a labeled diagram of a Cat6 Ethernet cable, showing the color order of wires in the T568B standard and labeling the RJ-45 connector pins.**

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2. **List three types of network cables (such as Cat5e, Cat6, and fiber optic) and for each, mention one real-world app or device (like a WiFi router, Zomato POS terminal, or IPL streaming server) where it would be used, explaining why.**

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Network Cable Type	Real-World Device/App Example	Why It Is Used
1. Cat5e Ethernet Cable	Wi-Fi router connected to a computer	It provides a reliable wired connection and supports speeds up to 1 Gbps over suitable distances.
2. Cat6 Ethernet Cable	Zomato restaurant POS terminal	It provides a fast, stable wired connection for processing orders and updating billing information.
3. Fiber Optic Cable	IPL live-streaming server network	It supports very high data speeds over long distances with low signal loss, making it suitable for transmitting large amounts of video data.

3. **Watch a YouTube video showing how to crimp an RJ-45 connector onto a network cable, then write down the step-by-step process in your own words and mention two common mistakes to avoid.**

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Materials required

- **Cat5e or Cat6 Ethernet cable**
- **RJ-45 connector**
- **Crimping tool**
- **Cable stripper/cutter**
- **LAN cable tester**

Step-by-step process

1. **Cut the cable: Cut the Ethernet cable to the required length using a cable cutter.**
2. **Strip the outer jacket: Remove about 2–3 cm of the outer covering carefully without damaging the inner wires.**
3. **Separate the wires: Untwist the four pairs and straighten the eight individual wires.**
4. **Arrange the wires: Follow the T568B colour order:**

White-Orange, Orange, White-Green, Blue, White-Blue, Green, White-Brown, Brown.

5. **Trim the wires: Cut all eight wires evenly so they fit properly inside the connector.**
6. **Insert into the RJ-45 connector: Insert the wires in the correct order until they reach the front of the connector. Ensure the outer cable jacket extends inside the connector.**
7. **Crimp the connector: Place the connector into the crimping tool and press firmly to secure the wires and connector.**
8. **Test the cable: Use a LAN cable tester to check that all eight pins are connected correctly.**

Two common mistakes to avoid

1. **Incorrect wire order: Mixing up the colour sequence can cause the cable to fail. Always check the T568B standard before crimping.**
2. **Loose or short wires: If the wires are not fully inserted or the outer jacket is not secured, the connection may become unreliable.**

4. **Given the following scenario: You are setting up a gaming zone for an IPL screening event and need to connect multiple screens and consoles. Choose the appropriate cable and connector types for high-speed, reliable connections, and explain your choices.**

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For an IPL screening event, I would choose the following cables and connectors to provide high-speed, reliable connections between screens, gaming consoles, and the network.

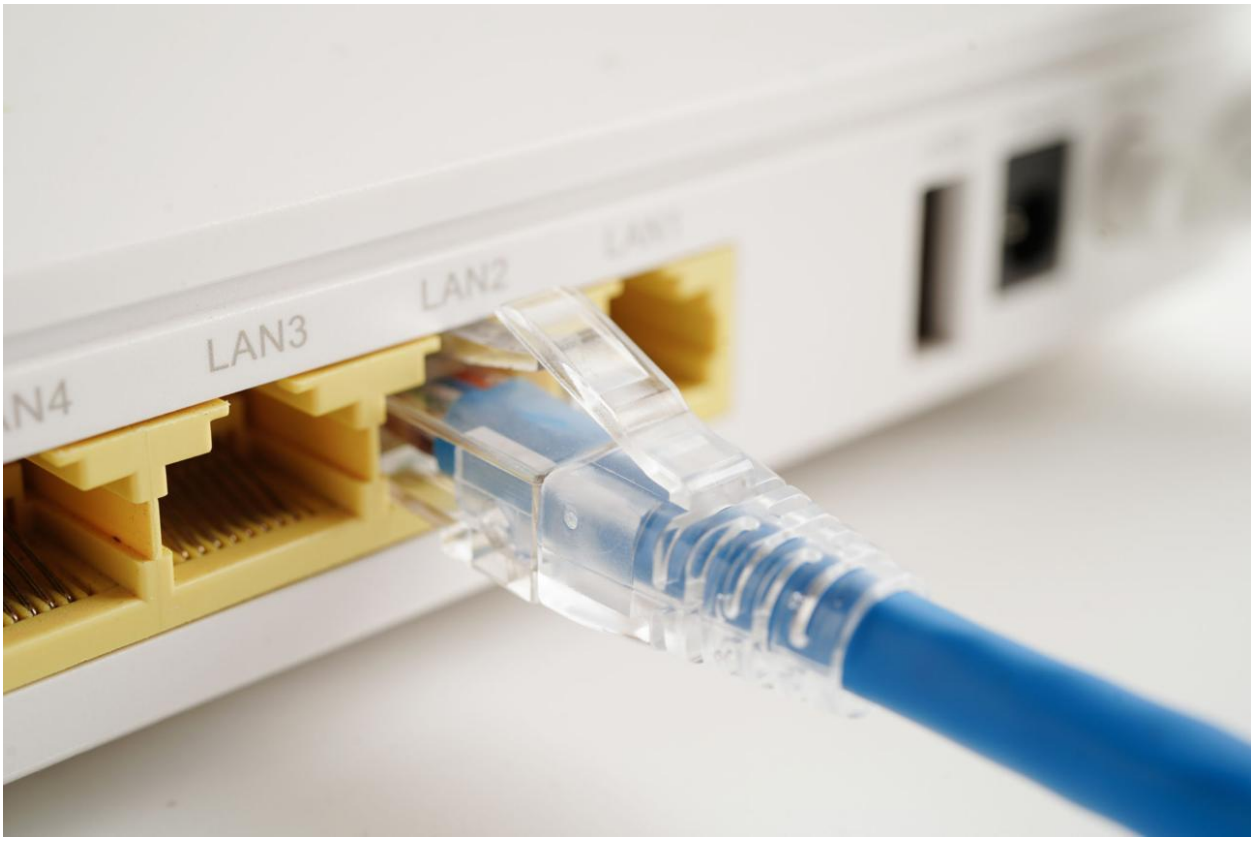
Cable type	Connector type	Purpose
HDMI 2.1 cable	HDMI connector	Connect gaming consoles to TVs or monitors for high-quality video and audio with low latency.
Cat6 Ethernet cable	RJ-45 connector	Connect consoles, PCs, and network switches for fast, stable internet and online gaming.
Fiber-optic cable	LC or SC connector	Connect network equipment over long distances with high speed and minimal signal loss.

Explanation

1. **HDMI 2.1: It supports high-resolution video and high refresh rates on compatible devices, making it suitable for gaming consoles and large screens.**
2. **Cat6 Ethernet: It supports up to 1 Gbps over standard 100-metre Ethernet links and provides a more stable connection than Wi-Fi for online gaming.**
3. **Fiber-optic cable: It is useful when the network switch and main server are far apart because it supports high-speed data transmission over longer distances.**

5. **Find and share a photo (from your home, college, or online) of any network cable or connector in use (such as behind a WiFi router or at a café's POS counter), and describe what type it is and what devices it connects.**
   Hint: Look for blue or yellow cables with clear plastic ends (RJ-45) or thin round cables for fiber optics.

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- **Cable type: Ethernet cable (Cat5e or Cat6; the exact category cannot be confirmed from the photo alone).**
- **Connector type: RJ-45 connector.**
- **Devices connected: The router and a wired network device, such as a computer, switch, or another networking device.**
- **Purpose: It transmits data between devices and provides a stable wired network connection. Ethernet connections are generally more reliable and have lower latency than Wi-Fi.**